

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of M. Sc. Backlog Examination , March, 2018

Course code & Name MS 712 & Biochemistry

Date: 27/03/2018 Day: Tuesday Time: 10.00 AM To 10.30 AM Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer/fill in the blanks and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		20
1.	Which of the following statements about the regulation of a metabolic pathway is correct?		01
	(i) Most metabolic pathways are regulated.	(ii) Most metabolic pathways are not regulated.	
	(iii) Metabolic regulation always depends on control by hormones.	(iv) Regulation of metabolic pathways always involves changing the amount of enzymes.	
2.	In the first step of glycolysis, the enzyme hexokinase uses ATP to transfer a phosphate to glucose to form glucose-6-phosphate. The product continues to be oxidized forming pyruvate in glycolysis and is a precursor to acetyl-CoA for the citric acid cycle. Suppose that a cell has only glucose available for energy and that the activity of hexokinase is suddenly stopped in this cell. Which of the following conditions will occur?		01
	(i) The cell will continue to produce energy from mitochondrial electron transport.	(ii) The cell will continue to produce ATP using the citric acid cycle.	
	(iii) The cell will ultimately be unable to produce ATP.	(iv) The cell will be forced to switch to fermentation to produce ATP.	

3.	L-lysine is an essential amino acid and must be supplied in the diet. By comparison, the stereoisomer D-lysine is not biologically active. Why can your body only utilize one form?		01
	(i) Since the L form and D form are enantiomers, they will bind together and inhibit utilization of the D form.	(ii) Since the L form and D form are chiral molecules, they will bind together and inhibit utilization of the D form.	
	(iii) Enzymes can only recognize a single, specific stereoisomer.	(iv) Antibodies recognize the D form and destroy it before your body can use it for nutritional purposes.	
4.	Almost all of the oxygen (O ₂) one consumes in breathing is converted to:		01
	(i) Water	(ii) Acetyl-CoA	
	(iii) Carbon dioxide	(iv) Carbon monoxide and then to carbon dioxide	
5.	A 7-year old girl presents with altered mental status, heart failure, and muscle weakness. Her serum level of ketones and glucose are abnormally low. She is diagnosed with primary carnitine deficiency. In which of the following is carnitine directly involved?		01
	(i) Activation of fatty acids	(ii) Transport of fatty acyl CoA	
	(iii) cytosol	(iv) β -Oxidation	
6.	In the absence of oxygen, the primary purpose of fermentation is to:		01
	(i) produce amino acids for protein synthesis	(ii) regenerate NAD ⁺ from NADH allowing glycolysis to continue	
	(iii) generate a proton gradient for ATP synthesis	(iv) oxidize glucose to generate reduce electron carriers	
7.	Which of the following compounds cannot serve as the starting material for the synthesis of glucose via gluconeogenesis?		01
	(i) Acetate	(ii) Propionate	
	(iii) Glycerol	(iv) Oxaloacetate	
8.	A modified form of cellulose found in the exoskeletons of insects and crustaceans is known as:		01
	(i) starch	(ii) chitin	
	(iii) amylose	(iv) glycogen	
9.	During the formation of the peptide bond which of the following takes place?		01
	(i) Hydrogen atom is lost from its carboxyl group of one amino acid and a hydroxyl group is lost from its amino group of another amino acid.	(ii) Hydroxyl group is lost from its carboxyl group of one amino acid and a hydrogen atom is lost from its amino group of another amino acid.	
	(iii) Hydroxyl group is lost from its carboxyl group of one amino acid and a hydroxyl group is lost from its amino group of another amino acid.	(iv) Hydrogen atom is lost from its carboxyl group of one amino acid and a hydrogen atom is lost from its amino group of another amino acid.	
10.	The rate of a second order reaction depends on the concentration of _____.		01
	(i) one substrate	(ii) two substrates	
	(iii) three substrates	(iv) none of the above	
11.	During oxidative phosphorylation, proton motive force that is generated by electron transport:		01
	(i) creates a pore in the inner	(ii) generates the substrates (ADP and Pi)	

	mitochondrial membrane.	for the ATP synthase.	
	(iii) induces a conformational change in the ATP synthase.	(iv) oxidizes NADH to NAD ⁺ .	
12.	You have recently identified a novel protein. After obtaining the protein sequence you realize that this protein contains several motifs and domains. How might this information help you to determine the function of your new protein?		01
	(i) All motifs and domains have a catalogued structure and function. Therefore, identification of such structures in the new protein will easily determine its function.	(ii) Motifs and domains are patterns that exist in protein structure and are often associated with particular functions. Therefore, identification of such sites can provide insight into possible functions of the unknown protein.	
	(iii) This information would not likely aid in determining the function of the new protein.	(iv) None of the above.	
13.	Which of the following coenzymes functions in acyl transfer reactions?		01
	(i) NADPH	(ii) Coenzyme A	
	(iii) biotin	(iv) None of the above.	
14.	Membrane proteins are not very soluble in water, because they possess long stretches of non-polar amino acids that		01
	(i) are too long to interact with the water molecules.	(ii) are transmembranal.	
	(iii) are hydrophobic.	(iv) are hydrophilic.	
15.	Which of the following is true of enzymes that are regulated by allosteric regulation?		01
	(i) They are usually multimeric enzymes.	(ii) A hyperbolic curve results from the plotting of V ₀ s [S].	
	(iii) They behave according to Michaelis-Menton kinetics.	(iv) None of the above.	
16.	The ω -Oxidation of Fatty Acids occurs in the _____.		01
17.	The key regulatory enzyme of fatty acid synthesis is _____.		01
18.	Enzyme cofactors that bind covalently at the active site of an enzyme are referred to as _____.		01
19.	A reversible inhibitor that can bind to either E alone or the ES complex is referred to as a _____.		01
20.	The number of double bonds in arachidonic acids are _____.		01

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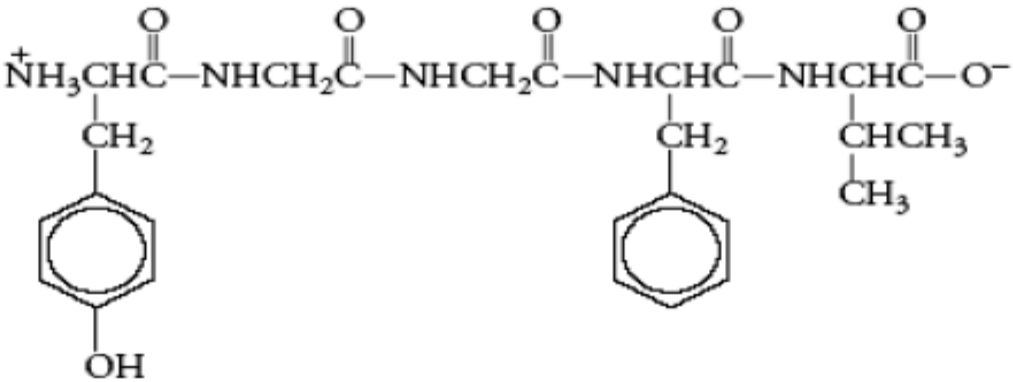
Course code & Name MS 712 & Biochemistry

Date: 27/03/2018 Day: Tuesday Time: 10.31 AM To 1.00 PM Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q - II Answer the following questions as directed (Any 5)		20
1.	Since both glycolysis and gluconeogenesis are irreversible, there is no thermodynamic barrier to their simultaneous operation. We know that gluconeogenesis is not simply a reversal of glycolysis. Why is this so? In your explanation be sure to mention at least two of the important steps and the enzymes involved.	04
2.	Calculate the net number of ATPs produced when one 18-carbon fatty acid salt is activated, enters the mitochondrion, and undergoes complete β oxidation. Include the ATP formed from acetyl-CoA in the citric acid cycle, and ATP from all of the reduced coenzymes that are produced.	04
3.	“Plants having sulpholipids rather than phospholipid.” justify.	04
4.	How do you sequence a protein? Discuss briefly.	04
5.	Can we predict the tertiary structure of a protein if we know its amino acid sequence? Justify your answer.	04
6.	Why we can't store glucose in the liver in its monomeric form? Justify your answer.	04
7.	Name the three ketone bodies. Are they all ketones?	04
Q-III Answer the following questions as directed (Any 10)		30
1.	“All enzymes are proteins but all proteins are not having enzymatic activity” Justify.	03
2.	What are the difference between transamination and oxidative deamination?	03
3.	What is lipoprotein? List out the types of lipoprotein and write their function.	03

4.	Where are the control points in the TCA cycle?	3
5.	Circle the peptide bonds and draw stars above the chiral carbons in the peptide below: 	03
6.	What is the source of NH_4^+ produced during amino acid catabolism? Oxidative deamination of glutamate produces which compound besides NH_4^+?	03
7.	Write short notes on Artificial Sweeteners	03
8.	What are the interactions that are involved in protein tertiary Structure?	03
9.	Write a short note on sugar alcohols.	03
10.	What are essential fatty acids? Give names of 3 essential fatty acids? What are the characteristics of sterol?	03
11.	What are glycosides? Give examples and functions of two glycosides.	03
12.	Give names of some Uncommon Amino Acids and mention also their important functions.	03
13.	Why is it important to specify the three-dimensional structure of amino acids?	03